Kids Playhouse Build Guide

A weekend-by-weekend plan for a 120 × 120 cm wooden playhouse: an enclosed room with a serving window and counter, a railed front porch, and a gable roof. All measurements are metric and match the annotated drawing.

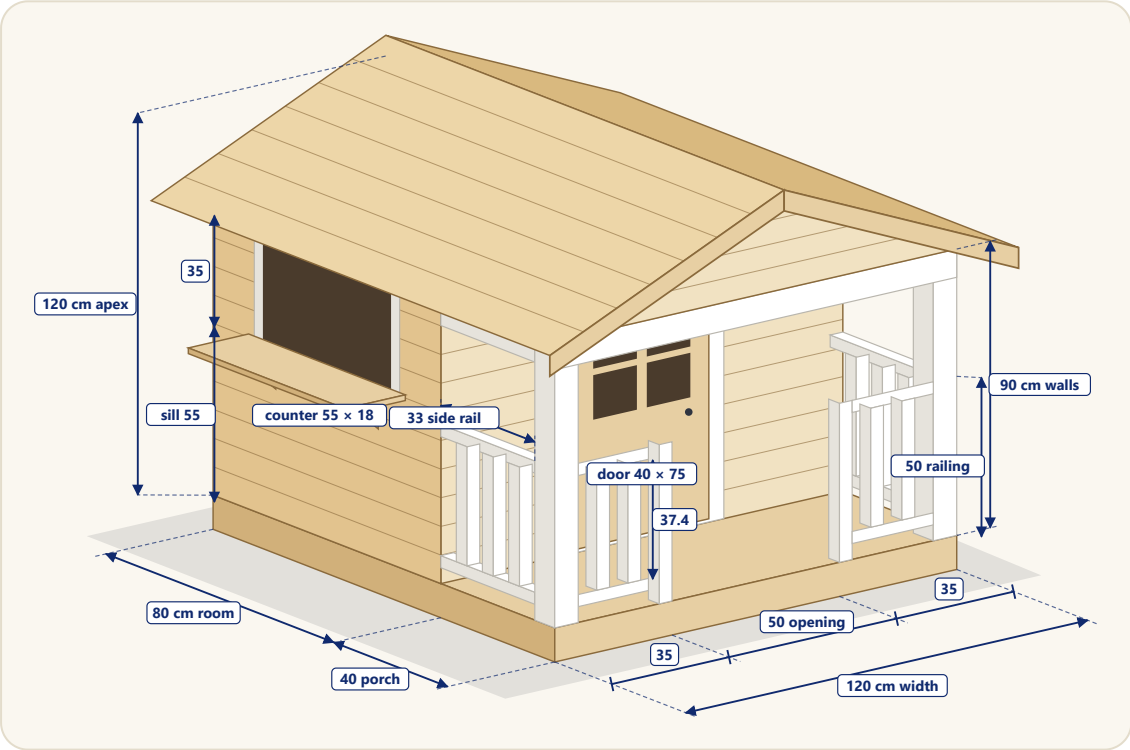
120 × 120 cm
Footprint (deck)

120 × 80 cm
Enclosed room (W × D)

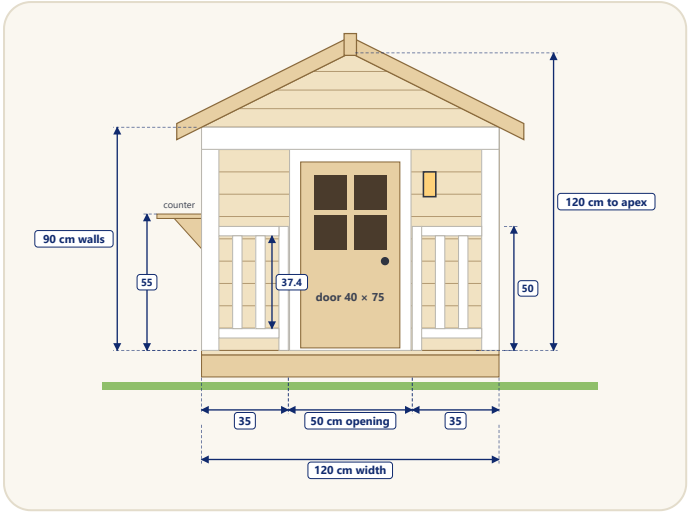
120 × 40 cm
Porch (W × D)

90 / 120 cm
Wall height / ridge height

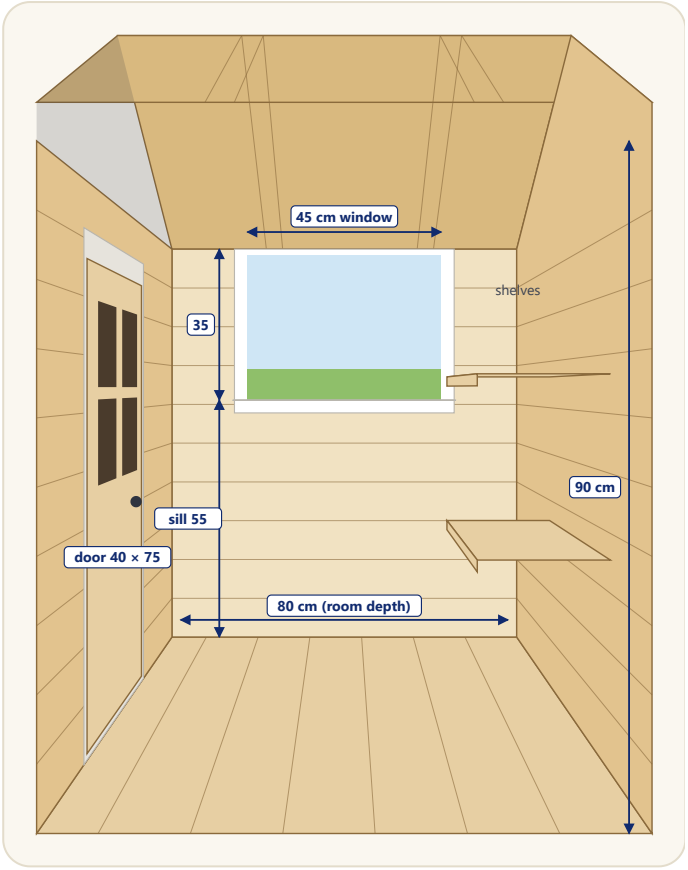
2 weekends
Typical build time, 1 to 2 people



Overview from the front left. The room is 120 × 80 cm, the porch in front of it is 120 × 40 cm with railings on three sides, and the serving window and counter are on the left side wall.



Front view. The porch pillars and railings stand in front of the door. From the front you only see the end of the counter.



Interior view, facing the serving window. The door wall is on your left and the back wall with shelves is on your right.

1. Key dimensions

ELEMENT	SIZE	NOTES
Deck / floor	120 × 120 cm	Covers both the room and the porch. 18 mm plywood on a 2×4 frame.
Enclosed room	120 W × 80 D	Back wall and two side walls are solid. Front wall has the door.
Porch	120 W × 40 D	Railings on the front and both sides, with a 50 cm opening in the middle of the front.
Wall height (eave)	90 cm	Top plate sits at 90 cm; this is also the top of the window.
Ridge height	120 cm	Roof rise 30 cm over a 60 cm run, about 27° (1:2 pitch).
Roof overhang	10 cm	All four sides. Ridge board length 140 cm.
Serving window	45 W × 35 H	Sill at 55 cm, top at 90 cm, centred on the left side wall.
Counter	55 × 18 cm	Outside, under the window at 55 cm, on two brackets.
Door	40 W × 75 H	Rough opening 45 × 80 cm, centred on the front wall. Four small panes in the top half.
Porch railing	50 cm high	Left 35 cm, opening 50 cm, right 35 cm. Baluster gaps no more than 9 cm.
Porch posts	7 × 7 cm, 90 cm	Two front corner posts carry the roof's front beam.

> The 80 / 40 cm split and the 35 / 50 / 35 railing split were scaled from the drawing. If you change them, keep the total depth at 120 cm so the roof and floor cut list still work.

2. Materials

ITEM	QTY	USE
Treated 2×4 (38 × 89 mm), 2.4 m	3	Floor frame, ridge, front beam, porch side beams
2×2 (38 × 38 mm), 2.4 m	16	Wall studs and plates, rafters, railings, balusters, counter brackets
Post 70 × 70 mm, 1.0 m	2	Porch corner posts
Plywood 18 mm, 1220 × 2440	1	Floor (1220 × 1220), door, counter top, interior shelves
Plywood 9 to 12 mm exterior, 1220 × 2440 (or tongue-and-groove cladding, about 9 m^2)	3	Wall cladding, gable triangles
Plywood 9 mm, 1220 × 2440	2	Roof sheathing, two panels 140 × 80 cm
Roofing felt or corrugated roofing, 3 m^2	1 roll / 2 sheets	Weatherproof roof cover
Concrete pavers or treated 4×4 skids	4 to 6	Keep the floor frame off the ground
Exterior screws 4 × 50 mm	1 box (200)	Framing
Exterior screws 4 × 30 mm	1 box (200)	Cladding, roof sheathing, trim
Exterior screws 5 × 80 mm	20	Floor frame, posts, beams
Metal angle brackets 50 mm	12	Posts to deck, rafters to plates
Door hinges 60 mm	2	Door
Door knob, magnetic catch	1 each	No locking latch. See safety.
Clear acrylic 3 mm (optional)	1 sheet 50 × 40 cm	Door panes. Leave the serving window open.

ITEM	QTY	USE
White exterior paint, child-safe wood stain or oil	1 L each	Trim white, walls natural
Exterior wood glue, sandpaper 80 / 120 / 180	1 each	Joints and finishing
Battery LED lantern (optional)	1	Porch light. No mains wiring.

3. Cut list

Lengths are nominal. Measure the real frame before cutting cladding, rafters and railing parts.

Floor frame (2×4)

PART	LENGTH	QTY
Rim joists (front and back)	120 cm	2
Joists (between rims)	112.4 cm	4
Floor sheet (18 mm ply)	120 × 120 cm	1

Walls (2×2, studs at 40 cm centres)

PART	LENGTH	QTY
Side wall plates (left and right, full 80 cm)	80 cm	4
Back and front wall plates (fit between the side walls)	112.4 cm	4
Studs (90 cm wall minus two 3.8 cm plates): right 3, left 4 (incl. 2 window studs), back 4, front 2	82.4 cm	13
Window sill (fits between the window studs, top face at 55 cm)	45 cm	1
Window cripples (bottom plate to sill)	47.4 cm	2
Door header (sits on the jack studs, underside at 80 cm)	52.6 cm	1
Door jack studs (bottom plate to header)	76.2 cm	2

PART	LENGTH	QTY
Packers (header to top plate)	2.4 cm	2

Porch (2×4 and posts)

PART	LENGTH	QTY
Corner posts (70 × 70), 3.8 × 8.9 cm notch in the front face of the top	90 cm	2
Front beam (2×4, sits in the post notches)	120 cm	1
Side beams (2×4, front wall to back face of post)	33 cm, cut to fit	2

Roof (2×2 rafters, 9 mm ply)

PART	LENGTH	QTY
Ridge board (2×4)	140 cm	1
Rafters, 27° plumb cut at the top, birdsmouth at the wall	78 cm	10 (5 pairs at 35 cm)
Roof panels	140 × 80 cm	2
Gable triangles (12 mm ply): 120 cm base, 30 cm high	—	2
Fascia and barge boards (2×2 or 1×4)	80 cm and 140 cm	4 + 2

Railings (2×2)

PART	LENGTH	QTY
Intermediate posts (35 cm in from each corner post)	50 cm	2
Top and bottom rails (corner post to intermediate post)	about 24 cm, cut to fit	4
Balusters, front (two per section, gaps about 5.5 cm)	37.4 cm	4
Side rails (side wall to corner pillar, both sides)	about 33 cm, cut to fit	4
Balusters, sides (three per side, gaps about 5.4 cm)	37.4 cm	6

Window, counter and door

PART	SIZE	QTY
Counter top (18 mm ply)	55 × 18 cm	1
Counter brackets (2×2 triangles)	15 × 15 cm	2
Window trim (1×3 or ply strips, painted white)	frame to fit 45 × 35 cm	1 set
Door (18 mm ply)	40 × 75 cm	1
Door pane cut-outs (2 × 2 grid)	4 × (12 × 14 cm)	1 set

4. Tools

Essential

- Cordless drill and driver with a 3 mm pilot bit and PZ2 or PH2 bits
- Circular saw or hand saw (a mitre saw speeds up the 27° cuts)
- Jigsaw for the window, door panes and gable triangles
- Tape measure, pencil, carpenter's square, spirit level
- Clamps (at least 2), sawhorses or a flat bench
- Sanding block or orbital sander
- Safety glasses, ear protection, dust mask, gloves

Nice to have

- Speed square for marking rafter angles
- Countersink bit so screw heads sit flush
- Router with a round-over bit for child-safe edges
- Staple gun for roofing felt
- Paint sprayer, or a 50 mm brush and small roller

5. Build steps

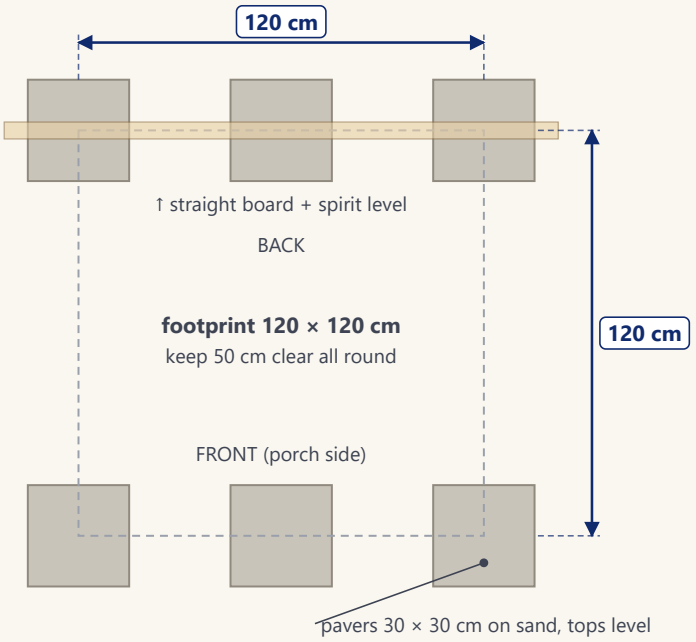
Each step is broken into small illustrated parts. Jump straight to the part you are building:

Parts you add in this picture Parts already built Opening or outline for reference

Dimension in cm

1 Plan the site and prepare the base

About 1 hour



Seen from above. Six pavers carry the corners and the middle of the front and back rim joists.

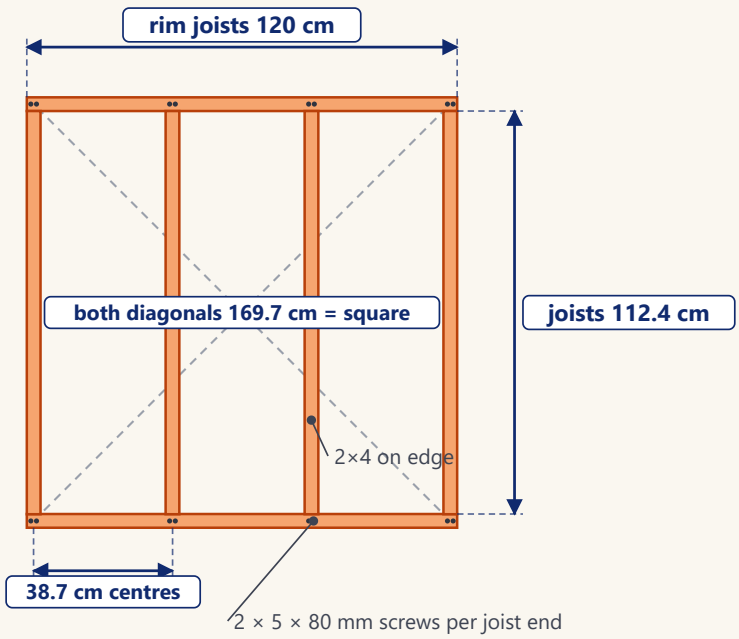
1a Set out and level the pavers

1. Choose a flat, well-drained spot with 50 cm of clear space around the 120 × 120 cm footprint.
2. Mark the footprint with four pegs and string. Check the diagonals are equal.
3. Dig out turf under each paver position, add 3 cm of sand, and bed a 30 × 30 cm paver at each corner and one in the middle of the front and back edges.
4. Lay a long straight board across the pavers in every direction and check with a spirit level. Add or scrape sand until all six tops are level.

> Decide now which side the window faces. The serving window is on the left side wall when you look at the front door.

2 Build the floor frame and deck

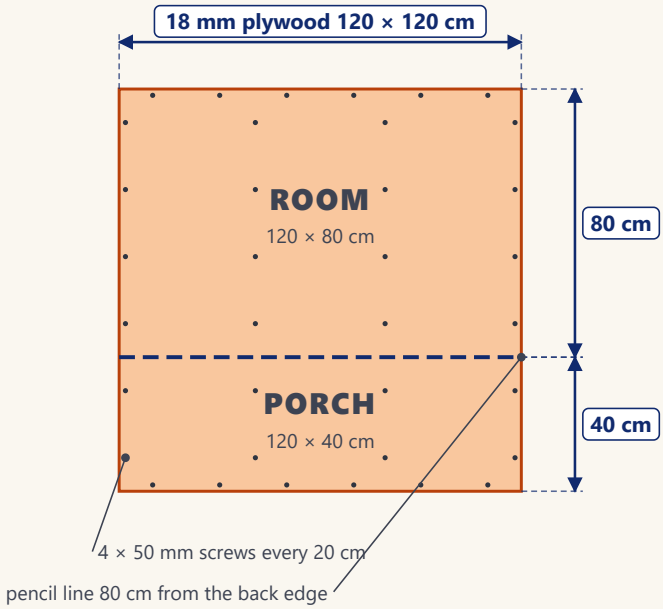
About 2 hours



Seen from above. Rim joists run left to right at the back and front; the four joists fit between them.

2a Assemble the 2×4 frame

1. Cut two rim joists at 120 cm and four joists at 112.4 cm from 2×4.
2. Stand all six pieces on edge on the pavers. Put a joist at each end of the rims and two in between, at about 38.7 cm centres.
3. Drill 3 mm pilot holes and drive two 5 × 80 mm screws through the rim into each joist end.
4. Square the frame: both diagonals must read the same, about 169.7 cm. Push the long corner in until they match.



The plywood covers the whole frame. The pencil line splits the room from the porch.

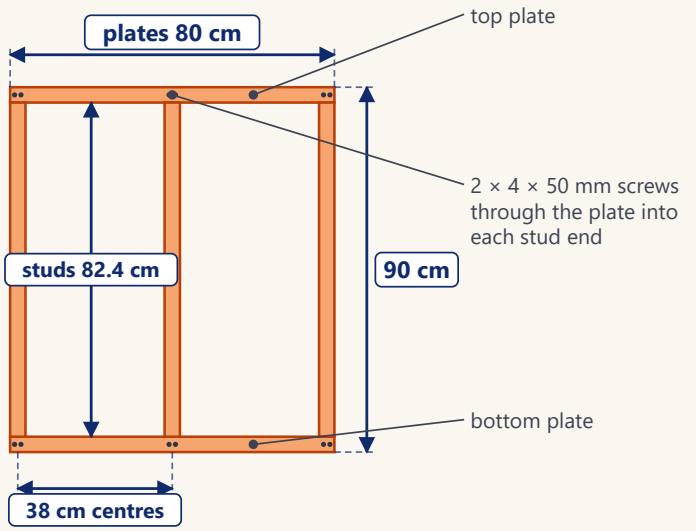
2b Fix the plywood deck

1. Cut the 18 mm plywood to 120 × 120 cm.
2. Lay it on the frame with all four edges flush. The sheet locks the frame square, so recheck the diagonals first.
3. Screw it down every 20 cm along every rim and joist with 4 × 50 mm screws. Snap a chalk line over each joist so you hit it.
4. Draw a pencil line across the deck 80 cm from the back edge. Everything in front of that line is porch.

3 Frame the four walls flat on the deck

About 3 hours

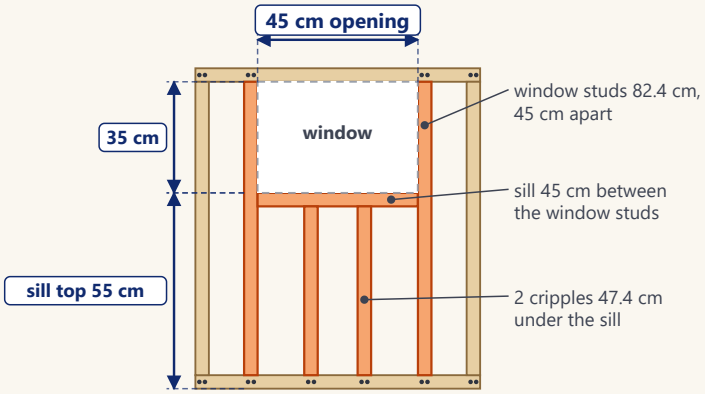
Build each wall lying flat on the deck, which is a flat, square work table. Every joint gets glue and two 4 × 50 mm screws driven through the plate into the stud end. All walls are 90 cm tall: 3.8 cm bottom plate, 82.4 cm studs, 3.8 cm top plate.



Right side wall seen from outside. This is the simplest wall, so build it first to get the method right.

3a Right side wall

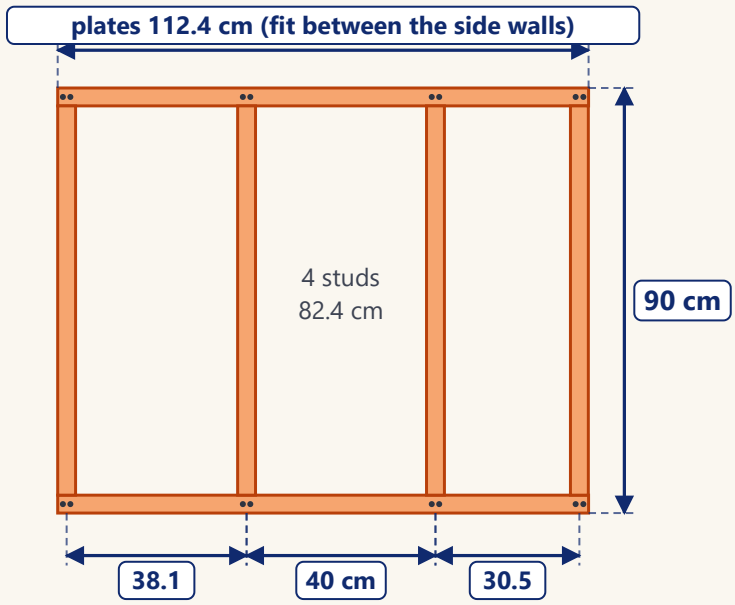
1. Cut two plates at 80 cm and three studs at 82.4 cm.
2. Lay the plates on edge, parallel, and mark stud positions: one flush at each end and one in the middle.
3. Glue and screw each stud between the plates.
4. Measure both diagonals and nudge the frame until they match, then fix a temporary diagonal batten across it to hold it square.



Left side wall seen from outside. The window runs from the sill up to the top plate.

3b Left side wall with the serving window

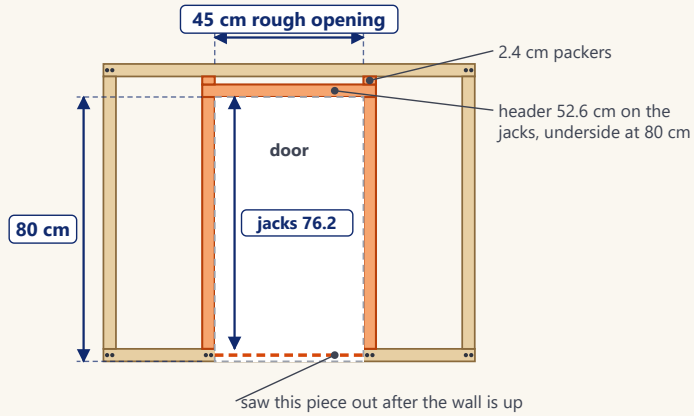
1. Cut two 80 cm plates, four 82.4 cm studs, one 45 cm sill and two 47.4 cm cripples.
2. Fix the two end studs. Fix the two window studs so their inner faces are 17.5 cm from each end, which leaves a 45 cm gap between them.
3. Screw the two cripples to the bottom plate inside the gap, about 15 cm in from each window stud.
4. Lay the 45 cm sill across the cripple tops, between the window studs. Its top face is at 55 cm. Screw through the window studs into the sill ends.
5. The top plate is the window head, so the opening is 45 × 35 cm.



Back wall seen from outside. It is 7.6 cm shorter than 120 because it fits between the side walls.

3c Back wall

1. Cut two plates at 112.4 cm and four studs at 82.4 cm.
2. Fix a stud flush at each end, then two more at 40 cm centres measured from the left end.
3. Square it by its diagonals and add a temporary batten.



Front wall seen from outside. The bottom plate stays in the opening until the wall is standing.

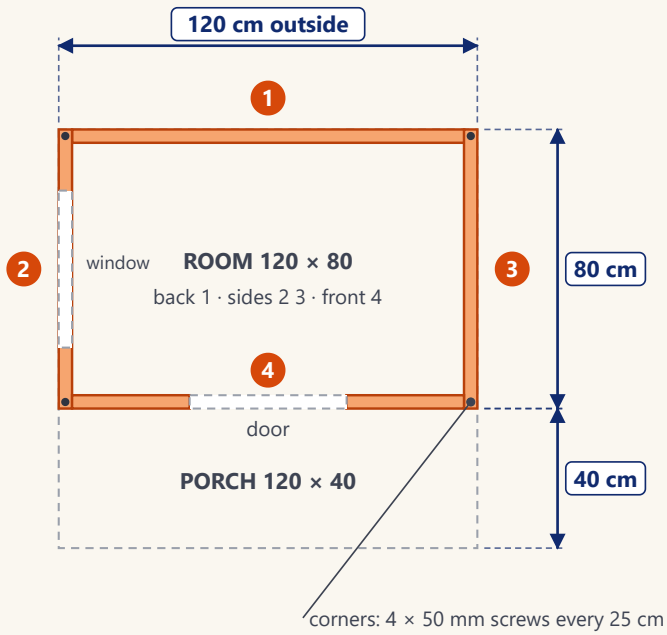
3d Front wall with the door opening

1. Cut two 112.4 cm plates, two 82.4 cm end studs, two 76.2 cm jack studs, one 52.6 cm header and two 2.4 cm packers.
2. Fix the end studs. Mark a 45 cm opening centred on the plates, 33.7 cm in from each end.
3. Fix a jack stud on each side of the opening, standing on the bottom plate.
4. Lay the header across the jack tops so its underside is at 80 cm. Fit a packer above each jack and screw down through the top plate into the header.

> Cut the bottom plate out of the door opening *after* the wall is standing, so the frame stays rigid while you move it.

4 Raise and join the walls

About 1.5 hours, two people



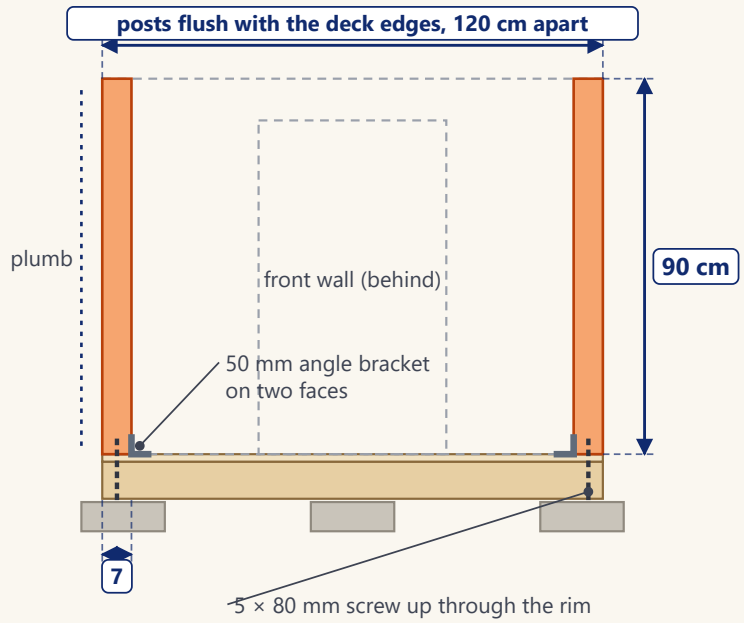
Seen from above at plate height. The numbers show the order to stand the walls.

4a Stand the walls in order

1. **Back wall (1):** stand it flush with the back edge of the deck and centred between the sides. Brace it with a scrap board, check plumb, and screw the bottom plate into the deck and rim joist every 30 cm with 5 × 80 mm screws.
2. **Left side wall (2):** stand it against the back wall's end, flush with the deck's left edge. Screw its end stud into the back wall's end stud every 25 cm with 4 × 50 mm screws.
3. **Right side wall (3):** repeat on the right.
4. **Front wall (4):** stand it between the side walls with its front face on the 80 cm pencil line. Screw its end studs to the side-wall studs and the bottom plate to the deck.
5. Check the outside measures 120 × 80 cm and the room's diagonals across the top plates are equal. Then remove the temporary battens.
6. Saw out the bottom plate across the door opening.

5 Set the porch pillars and beams

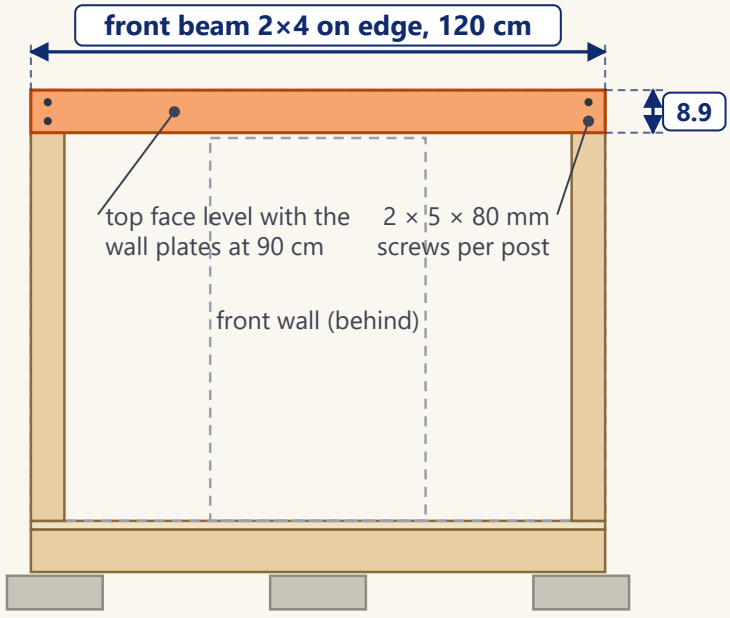
About 1.5 hours



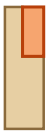
Seen from the front. The pillars stand at the two front corners of the deck.

5a Stand the corner pillars

1. Cut both 70 × 70 mm posts to exactly 90 cm, the same height as the walls.
2. Cut a notch 3.8 cm deep and 8.9 cm tall in the front face of each post top. This is where the front beam sits (see 5b).
3. Stand each post at a front corner, flush with the front and side edges of the deck.
4. Check it is plumb on two adjacent faces with a spirit level.
5. Fix two 50 mm angle brackets to the deck on the inner faces, then drive a 5 × 80 mm screw up from below through the rim joist into the post.
6. Add a temporary diagonal brace to hold it plumb.



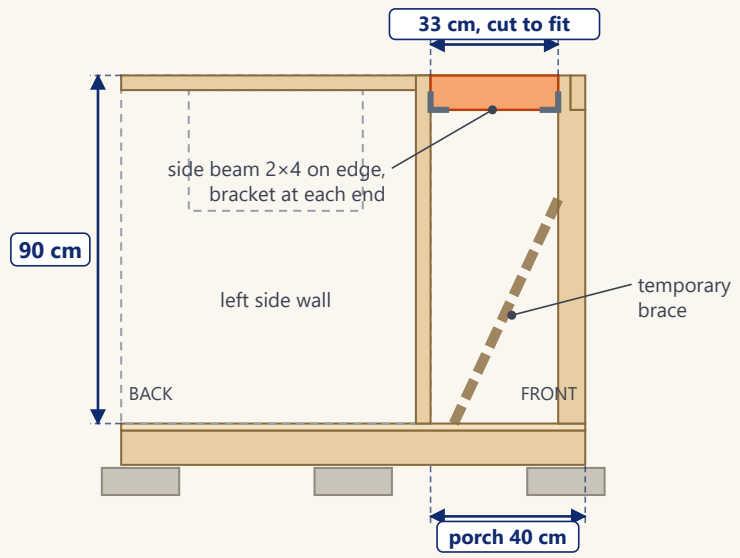
Notch each post top: 3.8 deep × 8.9 cm on the front face, so the beam sits flush. Side view: post (left), beam (orange)



Seen from the front. The beam's top face is level with the post tops and wall plates.

5b Fit the front beam

1. Cut a 2×4 at 120 cm.
2. Lift it on edge into the two post notches. Its ends are flush with the outer faces of the posts.
3. Lay a level across from the beam to the front wall's top plate. Both must be at 90 cm; plane or pack the notch if not.
4. Fix it with two 5 × 80 mm screws into each post.



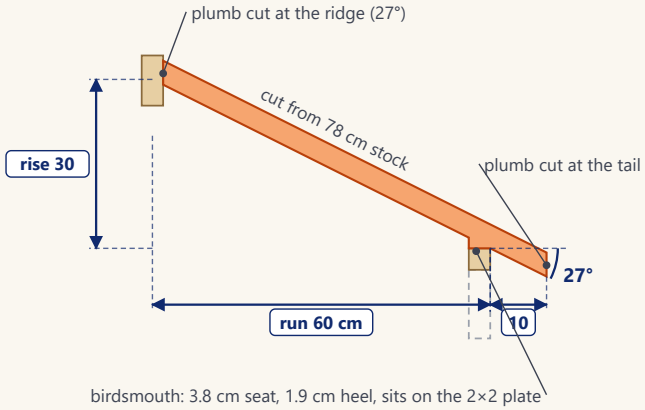
Seen from the left. The side beam ties the pillar back to the house.

5c Fit the two side beams

1. Measure from the front face of the side wall to the back face of the post, about 33 cm, and cut two pieces of 2×4.
2. Fit each one on edge in line with the side wall, top face at 90 cm, level with the side wall's top plate.
3. Fix it with a 50 mm angle bracket at each end: one into the side wall's end stud, one into the post.
4. Leave the temporary braces on until the roof is framed.

6 Frame the roof

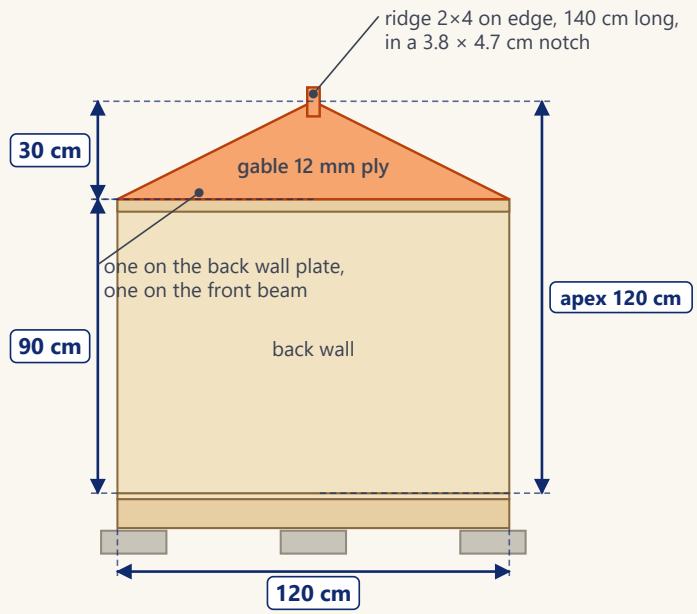
About 3 hours



One rafter seen from the gable end. Every rafter is a copy of this one.

6a Make a pattern rafter

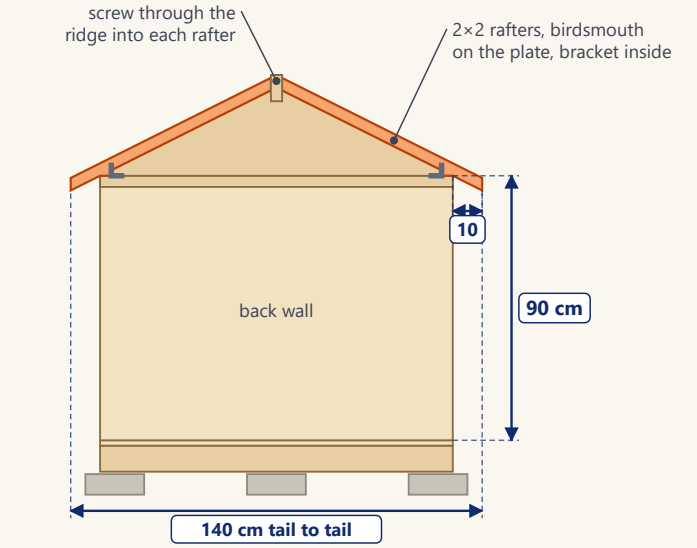
1. Take a 78 cm length of 2×2. Set a speed square or mitre saw to 27° (a 1:2 slope).
2. Make a plumb cut at the top end; this end butts against the ridge.
3. Measure 60.7 cm along the bottom edge from the top cut (54.3 cm level distance) and mark the birdsmouth: a 3.8 cm level seat and a 1.9 cm plumb heel, so it sits on the 2×2 top plate.
4. Cut the tail with a second plumb cut 10 cm beyond the outside of the wall.
5. Test it on both sides of the house against a scrap ridge held at the right height. When it fits, trace it onto nine more pieces.



The gable triangle at the back wall. An identical one sits on the porch front beam.

6b Fit the gables and the ridge

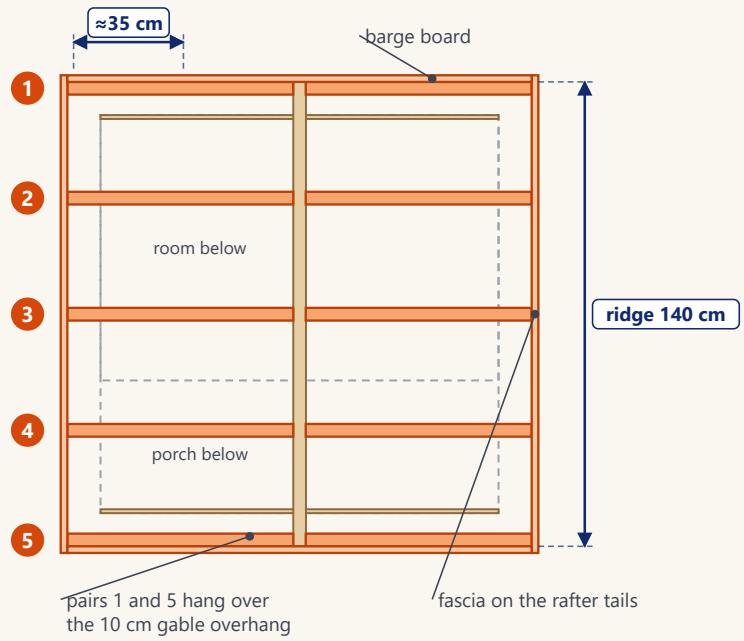
1. Cut two triangles from 12 mm ply: 120 cm base, 30 cm high, apex centred.
2. Cut a 3.8 cm wide, 4.7 cm deep notch at each apex for the ridge board.
3. Screw one triangle to the back wall's top plate and the other to the front beam, flush with their outer faces.
4. Cut the ridge from 2×4 at 140 cm. Drop it on edge into both notches so it overhangs 10 cm at the front and back. Screw through each gable into it.



A rafter pair in place. The birdsmouth seats on the plate and the top cut presses against the ridge.

6c Fix the first rafter pair

1. Hold a rafter with its top cut against the ridge and its birdsmouth on the wall plate.
2. Screw through the ridge into the rafter's top end with two 4 × 50 mm screws.
3. Fix the birdsmouth to the plate with a 50 mm angle bracket on the inside face.
4. Fix the matching rafter on the other side, opposite the first one.



The roof frame seen from above, before the sheathing goes on.

6d Fix all five pairs and the trim

1. Space five pairs along the ridge at about 35 cm centres. Pairs 1 and 5 are flush with the ridge ends, over the 10 cm overhang.
2. Pairs 2 and 3 sit on the side wall plates. Pair 4 sits on the porch side beams.
3. Screw a fascia board across the rafter tails on each eave.
4. Screw a barge board along the sloping edges of each gable end.
5. Remove the temporary post braces from step 5.

> Roof work is awkward on a 120 cm structure. Work from the ground on all four sides, not from the deck or a chair.

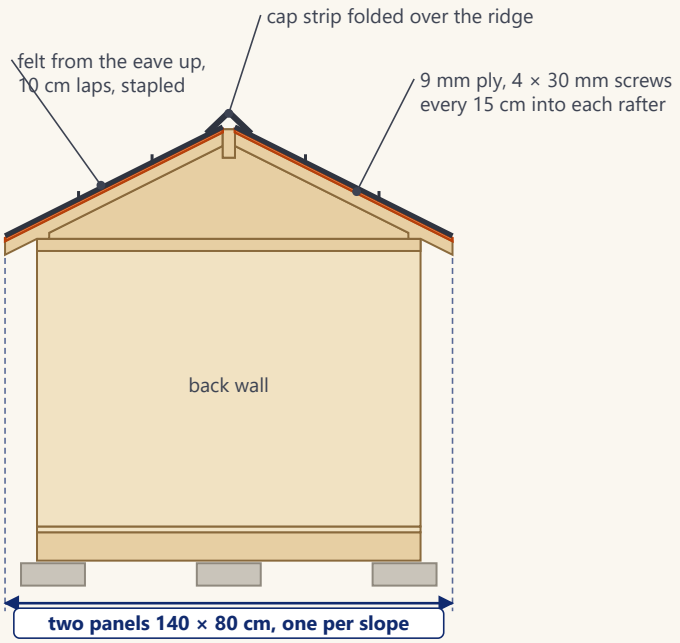
7 Clad the walls, fit the window and counter

About 3 hours

1. Cut cladding panels to the outside size of each wall (side walls 80 × 90 cm, back and front 120 × 90 cm). If using tongue-and-groove boards, run them horizontally and start at the bottom.
2. Screw cladding to studs and plates every 20 cm with 4 × 30 mm screws. Keep the bottom edge 5 mm above the deck so rain drains.
3. Mark and cut the window opening from outside with a jigsaw, following the frame: 45 × 35 cm with the bottom edge at 55 cm.
4. Mark and cut the door opening: 45 × 80 cm.
5. Frame the window with white trim strips, inside and out, so the cut plywood edge is covered.
6. Make the counter: 55 × 18 cm top, round the front corners, and fix it under the window on two triangular brackets screwed into the window studs. Top face at 55 cm.
7. The gable triangles from step 6 already close the roof ends. Paint or clad over them to match the walls.

8 Cover the roof

About 2 hours



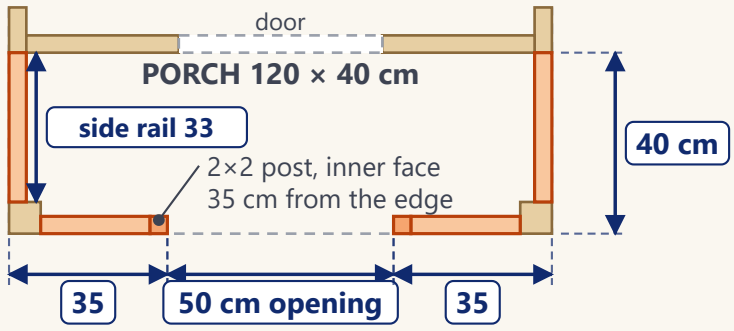
Seen from the gable end. The felt overlaps like shingles so rain runs off.

8a Sheath and weatherproof

1. Cut two 9 mm ply panels at 140 × 80 cm, one for each slope.
2. Lay each panel with its bottom edge flush with the fascia. Screw it to every rafter and the ridge with 4 × 30 mm screws every 15 cm.
3. Lay roofing felt from the eave upwards. Overlap each course by 10 cm and staple it.
4. Fold a cap strip over the ridge, overlapping both slopes, and nail it with felt nails.
5. Seal the ridge and the gable edges. Alternatively, fix corrugated roofing sheets with capped screws.
6. Check from inside for daylight.

9 Build the porch railings

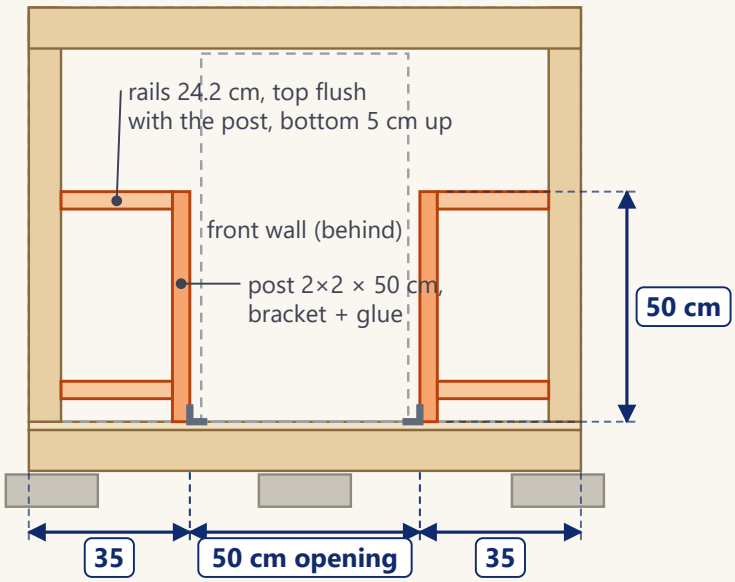
About 2 hours



The porch seen from above. Measure the intermediate posts from the outer edges of the deck.

9a Set out the railing posts

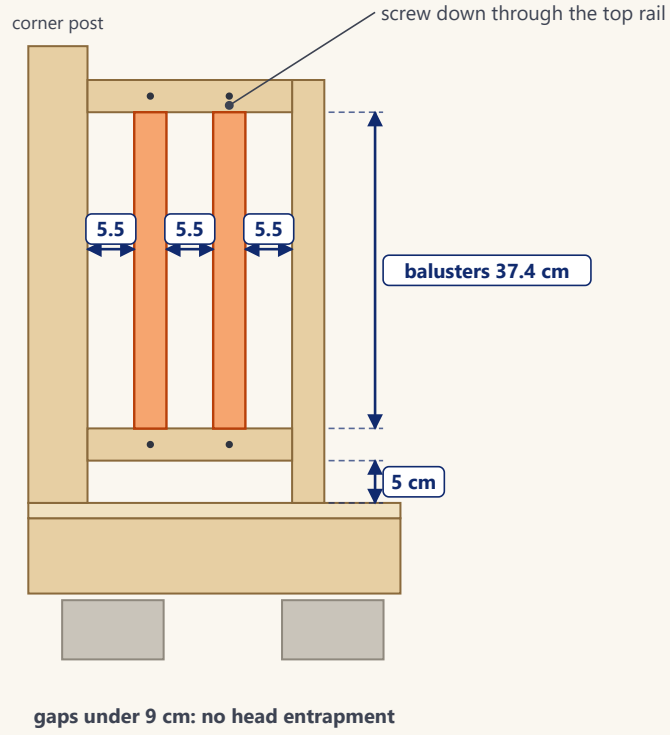
1. Along the front edge of the deck, mark 35 cm in from each side. These marks are the inner faces of the intermediate posts.
2. Check the gap between the two marks is 50 cm. That is the porch opening.



Seen from the front. Each railing section runs from a corner pillar to an intermediate post.

9b Fix the posts and rails

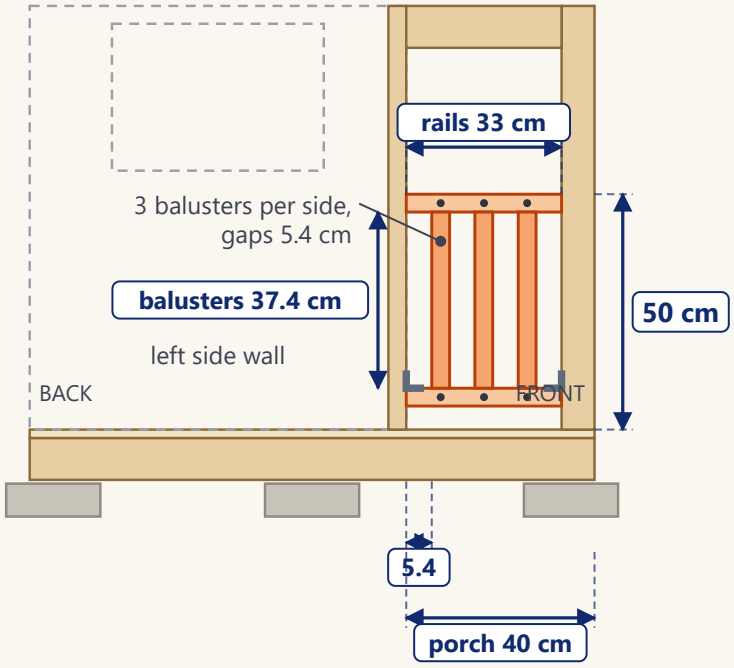
1. Cut two 2×2 posts at 50 cm. Glue each one on its mark, flush with the front edge, and fix it with an angle bracket on the opening side.
2. Measure between each corner pillar and its intermediate post, about 24 cm, and cut four rails.
3. Fix a bottom rail 5 cm above the deck and a top rail flush with the post top in each section, with glue and a bracket at each end.



One railing section up close. Both sections are the same.

9c Fit the balusters

1. Cut four 2×2 balusters at 37.4 cm.
2. Fit two per section between the rails, evenly spaced. The three gaps come out at about 5.5 cm, well under the 9 cm limit.
3. Screw down through the top rail and up through the bottom rail into each baluster end.
4. Round every top edge with sandpaper or a router. Cap the post tops with small square blocks if you like.



The left side of the porch seen from the side. The right side is the same, mirrored.

9d Close both porch sides

1. Measure from the front face of the side wall to the back face of the corner pillar, about 33 cm. Cut four rails to that length.
2. On each side, fix a bottom rail 5 cm above the deck and a top rail with its top at 50 cm. Keep their outer faces flush with the side wall and the pillar. Use glue and an angle bracket at each end.
3. Cut six balusters at 37.4 cm. Fit three per side between the rails, evenly spaced, which leaves four gaps of about 5.4 cm.
4. Screw through the rails into each baluster end and round every top edge.

10 Hang the door, finish and furnish

About 3 hours plus drying time

1. Cut the door to 40 × 75 cm from 18 mm ply. Cut four pane openings in a 2 × 2 grid in the upper half and glue a single piece of 3 mm acrylic behind them, or leave them open.
2. Hang the door on two hinges with a 5 mm gap all round. Fit a knob and a magnetic catch. Fit a soft finger guard strip on the hinge side.
3. Sand everything to 120 grit, then 180 on edges children touch. Round all corners.
4. Paint trim, posts, railings and the window frame white. Oil or stain the walls, floor and door with a child-safe exterior finish. Two coats, 24 hours apart.
5. Add the battery lantern beside the door, interior shelves from plywood offcuts, and a doormat.
6. Final check: push on every wall and post. Nothing should rack or wobble. Anchor the deck to the pavers or ground with brackets or pegs.

6. Safety checklist

- **Gaps:** any opening between railings, balusters or under rails must be under 9 cm, or over 23 cm, to avoid head entrapment.
- **Edges and fixings:** round all edges, countersink every screw, and check that no screw tips protrude inside or out.
- **Door:** no lock or latch that can trap a child inside. Use a magnetic catch and a finger guard.
- **Glazing:** use acrylic, never glass. The serving window stays open.
- **Finishes:** choose paint and stain labelled safe for children's toys or furniture, and let them cure fully before use.
- **Stability:** anchor the base and keep the playhouse away from slopes, fences children could climb over, and anything they could jump onto from the roof.
- **Ground:** grass, bark or rubber mats around the porch soften falls from the 15 cm deck step.
- **Maintenance:** each spring, check for splinters, loose railings and soft wood, and recoat the finish.

Dimensions follow the annotated reference drawing. Nominal lumber sizes are 2×2 = 38 × 38 mm and 2×4 = 38 × 89 mm. Measure your actual timber before cutting.